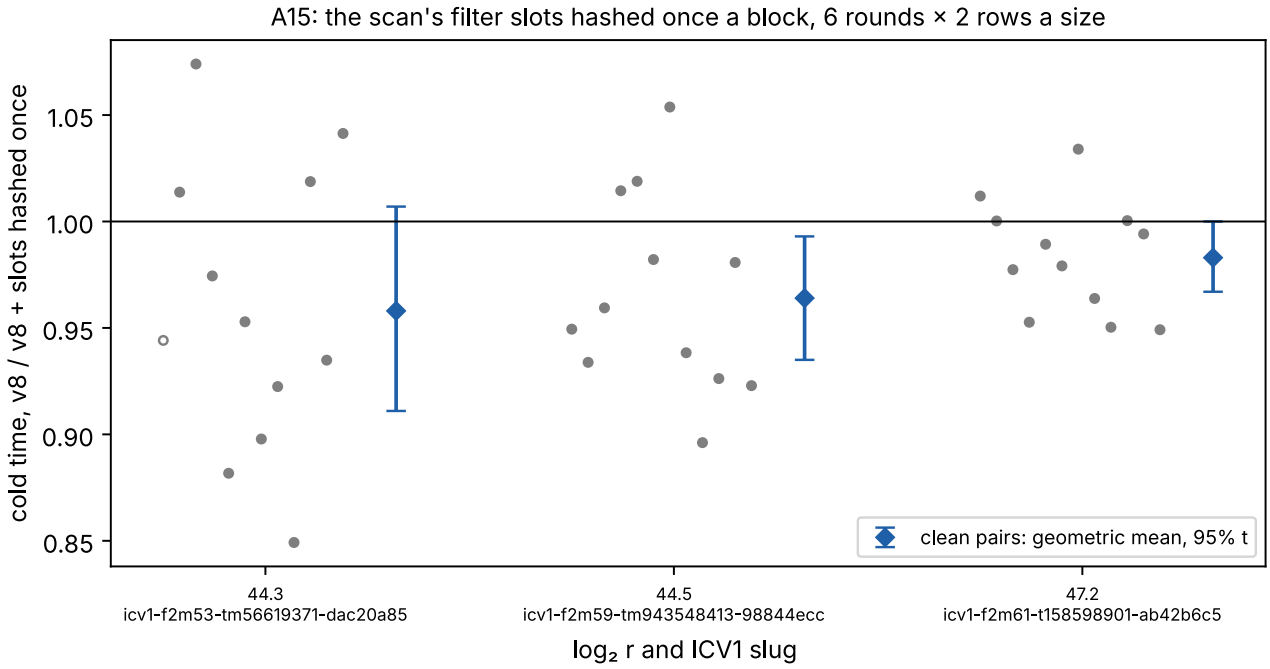
A15: each scanned key's filter slot hashed once a block

Exploration, negative, no round. v8 (`ic-r12-v7-23138d20`) against v8 with each key's filter word and bits computed once a block into two arrays (`ic-a15-v8-7ec4e6cb`, `filter-slots.patch`), instead of hashing each key twice (prefetch ahead, then test). Run 2026-10-17 03:50–04:08 UTC on the Emerald Rapids host R12 ran on: 72 isolated processes on CPU 2, six rounds with the order alternating, `M1`'s two rows at the three target sizes. Records: `research/ic_tool_program/explorations/A15-filter-slots-20261017/`.

- **Completed:** 72 of 72.
  - One `a15` run was contended and is excluded.
  - The busy gate refused 39 starts, and each refused run was asked for again.
- **Logarithms:** every pair recovered the same logarithm in both arms (36 of 36).
- **Outcome:** the candidate is 2–4% slower cold at every size, and collection is 2–6.5% slower. The filter stage is bound by its memory access, not by the hash.



research/ic_tool_program/explorations/A15-filter-slots-20261017/pairs.tsv; 72 isolated processes; open circles: a pair with a contended run

Figure 1: Cold time, v8 over the candidate, per pair (open circle: a contended run), with the clean pairs' geometric mean and 95% t interval. Values from `pairs.tsv` and `summary.tsv`.

Cold time, v8 / a15

curve	$\log_2$ r	clean pairs	cold	95%	all pairs
`icv1-f2m53-tm56619371-dac20a85`	44.3	11	0.958	[0.911, 1.007]	0.957
`icv1-f2m59-tm943548413-98844ecc`	44.5	12	0.964	[0.935, 0.993]	0.964
`icv1-f2m61-t158598901-ab42b6c5`	47.2	12	0.983	[0.967, 1.000]	0.983

Phases, v8 / a15 (geometric mean of 12 pairs' median repetitions)

curve	collection	build	selection
`icv1-f2m53-tm56619371-dac20a85`	0.935	0.985	1.045
`icv1-f2m59-tm943548413-98844ecc`	0.944	0.999	1.018
`icv1-f2m61-t158598901-ab42b6c5`	0.980	0.997	1.050

Median cold time over 12 processes, v8 → a15: 412.1 → 423.7 ms, 422.7 → 441.4 ms and 1647.7 → 1670.0 ms.

Scope

One host, one core, three sizes, two rows a size. No tests were run on the candidate's tree, and it is not landed.